COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE-010: Data Structures and Algorithm	**SECTION:**	CPE21S3
DATE PERFORMED:	10 / 20 / 26	**DATE SUBMITTED:**	10 / 20 / 26
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Hands-on Activity 14.1: Algorithm Complexity

1. PROCEDURE OUTPUT

	Best Case	Worst Case	Analysis
Merge Sort	**O(n log n)**	**O(n log n)**	Merge sort has a constant best case, average case, and worst case as the sort will always divide the array and does not matter if it is sorted or not.
Insertion Sort	**O(n)**	**O(n^2)**	The worst case of insertion sort is in reverse as it has to compare and replace past elements to sort the array. The average case is randomized, but the time complexity is still (O)n^2 because it still needs to compare elements with previous elements. The best case of the insertion sort is in order because all the sort has to do is compare the variables in each array, and it does not need to compare previous variables and sort them out.

Table 14-1. Best Case and Worst-Case Analysis using Theoretical Tools

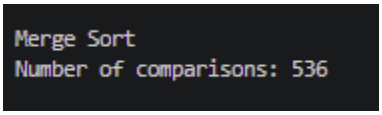
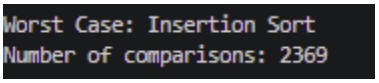
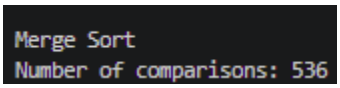
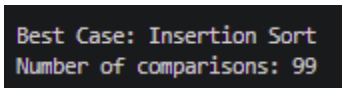
	Merge Sort	Insertion Sort
Best Case		
Worse Case		

Table 14-2. Best-Case and Worst-Case Analysis using Experimental Tools

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

ILO A: Contrast and compare the two searching techniques used in class. The required deliverable for this activity would be the theoretical and empirical analysis. For the theoretical analysis, the deliverable must be:

- **Pseudocode Analysis**

```
START

  DECLARE sizes = [1000, 10000, 100000]

  FOR each size in sizes DO

    n = size
    CREATE array of n integers

    // Initialize the array
    FOR i = 0 TO n - 1 DO
       arr[i] = i
    END FOR

    target = n

    // Binary Search
    start = current time
    PERFORM binary search on arr for target
    end = current time
    binaryTime = end - start

    // Linear Search
    start = current time

    FOR i = 0 TO n - 1 DO
```

```
        IF arr[i] == target THEN
            BREAK
        END IF
    END FOR

    end = current time
    linearTime = end - start

    // Display Results
    PRINT "Input Size: ", n
    PRINT "Binary Search: ", binaryTime, " seconds"
    PRINT "Linear Search: ", linearTime, " seconds"
    PRINT a blank line

  END FOR

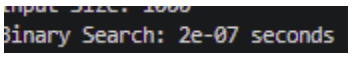
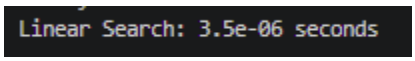
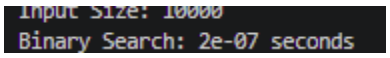
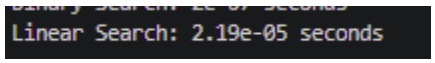
END
```

Analysis - The pseudocode declares an integer that determines the max size. It then creates a for loop where it creates an array based on the max size. Then, it gives random numbers in the array. A target is set for the searching algorithms to find. Then, it starts the binary and linear search algorithms, both of which have timers that determine the runtime of each searching algorithm on how long they did the operation.

- **Time and Space Complexities**
  Binary Search has a time complexity of T(n) and a space complexity of S(1) during the worst case.
  Linear Search has a time complexity of T(n) and a space complexity of S(1) during the worst case.

  Both worst case case scenarios are the same and since they both search through every index during the worst case scenario.

For the empirical analysis, the deliverable must be the filled-out table and your analysis.

Input Size (n)	Elapsed Time for Binary Search (Worst-Case)	Elapsed Time for Linear Search (Worst-Case)
1,000		
10,000		

100,000	Input Size: 100000 Binary Search: 2e-07 seconds	Linear Search: 0.0002099 seconds

Conclusion:

Fabonan - I conclude my learning of time and space complexity where different algorithms have different time and space complexities, where the best goal in algorithms is to shorten the time and size complexities in best case and worst case.

Santos - I have learned much about time and space complexity. I learned that it is difficult to reach an exponential best case scenario sorting as it can possibly take years or even centuries to finish. I learned that the goal in making an algorithm is to have the shortest time possible to organize data using different sorting methods or searching methods to reduce the complexity.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion
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